ADVANCED AND QUANTUM OPTICS

Model Solutions and Marking Scheme

Natural Sciences Tripos Part II — 2026

SECTION A

Question 1 [4 marks] — Gaussian Beam Parameters

The Rayleigh range is

$$z_0 = \frac{\pi W_0^2}{\lambda} = \frac{\pi(0.5 \times 10^{-3})^2}{633 \times 10^{-9}} = 1.24 \text{ m}\,.$$

[**1 mark**] for correct formula and value.

The beam divergence half-angle is

$$\theta_0 = \frac{W_0}{z_0} = \frac{\lambda}{\pi W_0} = \frac{633 \times 10^{-9}}{\pi \times 0.5 \times 10^{-3}} = 4.03 \times 10^{-4} \text{ rad} \approx 0.40 \text{ mrad}\,.$$

[**1 mark**] for divergence.

At distance $z = 10$ m from the waist, the beam radius is

$$W(z) = W_0\sqrt{1 + \left(\frac{z}{z_0}\right)^2} = 0.5 \times 10^{-3}\sqrt{1 + \left(\frac{10}{1.24}\right)^2}\,.$$

Since $z/z_0 = 8.06 \gg 1$, we have $W(z) \approx W_0 \cdot z/z_0 = 4.03$ mm. The beam diameter is $2W \approx 8.1$ mm.

[**2 marks**] for correct $W(z)$ calculation and final diameter.

Question 2 [4 marks] — Phase Matching in Nonlinear Optics

For sum-frequency generation with two input waves at ω_1 and ω_2:

Frequency-matching condition: $\omega_3 = \omega_1 + \omega_2$ (conservation of energy). [**1 mark**]

Phase-matching condition: $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3$ (conservation of momentum). [**1 mark**]

Physical explanation: The nonlinear polarisation at ω_3 acts as a source that radiates at all points along the crystal. For the generated wave to grow coherently, the phase of the driving polarisation must stay matched to the phase of the generated free wave. If $\Delta\mathbf{k} = \mathbf{k}_1 + \mathbf{k}_2 - \mathbf{k}_3 \neq 0$, contributions from different points along the crystal interfere destructively, and the generated power oscillates rather than growing monotonically. The coherence length is $l_c = \pi/|\Delta k|$. [**1 mark**]

Practical method: In a uniaxial birefringent crystal, the ordinary and extraordinary rays experience different refractive indices. By choosing the polarisations of the input and output waves appropriately (e.g. Type-I: both inputs ordinary, output extraordinary), one can find an angle of propagation relative to the optic axis at which $n_o(\omega_1) = n_o(\omega_2)$ and $n_e(\omega_3)$ satisfy $k_1 + k_2 = k_3$. This is called angle-tuned or critical phase matching. [**1 mark**]

Question 3 [4 marks] — Dielectric Waveguide Modes

The numerical aperture is

$$\mathrm{NA} = \sqrt{n_1^2 - n_2^2} = \sqrt{1.50^2 - 1.48^2} = \sqrt{2.25 - 2.1904} = \sqrt{0.0596} = 0.244\,.$$

[**2 marks**] for correct NA calculation.

The number of TE modes is

$$M = \left\lceil \frac{2d}{\lambda_0}\,\mathrm{NA} \right\rceil = \left\lceil \frac{2 \times 5.0 \times 10^{-6}}{1.55 \times 10^{-6}} \times 0.244 \right\rceil = \left\lceil \frac{10.0}{1.55} \times 0.244 \right\rceil = \lceil 1.57 \rceil = 2\,.$$

The waveguide supports **2 TE modes**. [**2 marks**] for correct formula and result. (Accept $M = 1$ if using the single-mode condition $M \approx 2d\,\mathrm{NA}/\lambda_0$ rounded down, with clear reasoning about the $m = 0$ mode always being allowed.)

Question 4 [4 marks] — Resonator Parameters

Free spectral range:

$$\nu_F = \frac{c}{2d} = \frac{3 \times 10^8}{2 \times 0.20} = 7.50 \times 10^8\ \mathrm{Hz} = 750\ \mathrm{MHz}\,.$$

[**1 mark**]

Mirror losses:

$$\alpha_{m1} = \frac{1}{2d} \ln \frac{1}{\mathcal{R}_1} = \frac{1}{0.40} \ln \frac{1}{0.99} = 0.0251\ \mathrm{m}^{-1}\,, \quad \alpha_{m2} = \frac{1}{2d} \ln \frac{1}{\mathcal{R}_2} = \frac{1}{0.40} \ln \frac{1}{0.97} = 0.0761\ \mathrm{m}^{-1}\,.$$

Total: $\alpha_r = \alpha_{m1} + \alpha_{m2} = 0.101\ \mathrm{m}^{-1}$.

Finesse:

$$\mathcal{F} = \frac{\pi}{\alpha_r d} = \frac{\pi}{0.101 \times 0.20} = 155\,.$$

Alternatively, $\mathcal{F} = \pi \sqrt[4]{\mathcal{R}_1 \mathcal{R}_2}\,/\,(1 - \sqrt{\mathcal{R}_1 \mathcal{R}_2}) = \pi(0.9604)^{1/4}/(1 - 0.9604^{1/2}) \approx 155$. [**1.5 marks**]

Photon lifetime:

$$\tau_p = \frac{1}{\alpha_r c} = \frac{1}{0.101 \times 3 \times 10^8} = 33.0\ \mathrm{ns}\,.$$

[**1.5 marks**]

SECTION B

Question 5 [17 marks] — Lasers and Spectral Properties

(a) [3 marks]

For an atom in a cavity of volume V with n photons in a mode of frequency ν:

Absorption (atom in level 1, n photons present):

$$P_{\text{ab}} = n\,\frac{c}{V}\,\sigma(\nu)\,.$$

Stimulated emission (atom in level 2, n photons present):

$$P_{\text{st}} = n\,\frac{c}{V}\,\sigma(\nu)\,.$$

Spontaneous emission (atom in level 2, independent of n):

$$P_{\text{sp}} = \frac{c}{V}\,\sigma(\nu)\,.$$

[**1 mark each**] for correct expressions. Note that absorption and stimulated emission have the same functional form $\propto n$, while spontaneous emission is independent of n.

(b) [4 marks]

Setting $dN_2/dt = 0$: $N_2 = R_2\tau_2$. **[0.5 marks]**

Setting $dN_1/dt = 0$: $N_1 = \tau_1\left(\frac{N_2}{\tau_{21}} - R_1\right) = \tau_1\left(\frac{R_2\tau_2}{\tau_{21}} - R_1\right)$. **[1 mark]**

Therefore

$$N_0 = N_2 - N_1 = R_2\tau_2 - \tau_1\left(\frac{R_2\tau_2}{\tau_{21}} - R_1\right) = R_2\tau_2\left(1 - \frac{\tau_1}{\tau_{21}}\right) + R_1\tau_1\,.$$

[1.5 marks]

Conditions for large N_0: [1 mark]

- Large pump rates R_1, R_2.
- Long upper-state lifetime τ_2 (so level 2 retains population).
- Short lower-state lifetime $\tau_1 \ll \tau_{21}$ (so level 1 empties quickly).
- $\tau_{21} \approx t_{\text{sp}}$ with $t_{\text{sp}} \gg \tau_1$ (fast depopulation of lower level relative to the radiative lifetime).

(c) [3 marks]

Including stimulated transitions, the rate equations become:

$$\frac{dN_2}{dt} = R_2 - \frac{N_2}{\tau_2} - (N_2 - N_1)W_i\,,$$
$$\frac{dN_1}{dt} = -R_1 - \frac{N_1}{\tau_1} + \frac{N_2}{\tau_{21}} + (N_2 - N_1)W_i\,.$$

Setting both time derivatives to zero and solving for $N = N_2 - N_1$ (subtracting the equations and using the expressions from part (b)):

$$N = \frac{N_0}{1 + \tau_s W_i} ,$$

where the saturation time constant is

$$\tau_s = \tau_2 + \tau_1 \left(1 - \frac{\tau_2}{\tau_{21}}\right) ,$$

which is always positive since $\tau_2 \leq \tau_{21}$. [**1.5 marks for derivation, 0.5 marks for τ_s**]

The saturation photon-flux density is defined by $\tau_s W_i = \phi/\phi_s$, giving

$$\phi_s(\nu) = \frac{1}{\tau_s \, \sigma(\nu)} .$$

[**1 mark**]

(d) [3 marks]

Gain condition: The small-signal gain must exceed the loss: $\gamma_0(\nu) > \alpha_r$, or equivalently $N_0 > N_t = \alpha_r/\sigma(\nu)$. [**0.5 marks**]

Phase condition: Round-trip phase must be a multiple of 2π: $2kd = 2\pi q$, giving resonant frequencies $\nu_q = q\,c/(2d)$. [**0.5 marks**]

Above threshold, equating saturated gain to loss:

$$\frac{\gamma_0(\nu)}{1 + \phi/\phi_s(\nu)} = \alpha_r .$$

Solving for ϕ:

$$\boxed{\phi = \phi_s(\nu) \left(\frac{\gamma_0(\nu)}{\alpha_r} - 1\right)}, \qquad \gamma_0(\nu) > \alpha_r .$$

Below threshold ($\gamma_0 \leq \alpha_r$), $\phi = 0$. [**2 marks**]

(e) [4 marks]

Homogeneously broadened medium: All atoms share the same lineshape. Gain saturation is uniform across all frequencies. As the strongest mode (closest to ν_0) depletes the population inversion, the gain drops below the loss for all other modes. Result: *single-mode operation* in steady state.

[**1 mark for description, 1 mark for sketch**] showing gain curve uniformly pulled down to the loss level, with one surviving mode.

Inhomogeneously broadened medium: Different groups of atoms contribute to different parts of the gain curve (e.g. Doppler broadening in a gas laser). Each mode saturates only those atoms whose resonance frequencies coincide with it, creating dips or "holes" in the gain profile. Multiple modes can therefore oscillate simultaneously, each sustained by its own subset of atoms.

[**1 mark for description, 0.5 marks for sketch**] showing spectral holes burned into the gain profile at each lasing mode.

Spectral hole burning is the process whereby a strong intracavity field at a particular frequency selectively depletes the inverted population of those atoms that are resonant at that frequency, creating a localised dip (hole) in the gain spectrum. [**0.5 marks**]

Question 6 [17 marks] — Coherence, Interferometry, and Nonlinear Optics

(a) [3 marks]

The **complex degree of temporal coherence** is defined as

$$g(\tau) = \frac{G(\tau)}{G(0)} = \frac{\langle U^*(t)\, U(t+\tau)\rangle}{\langle |U(t)|^2\rangle}\,,$$

where $G(\tau) = \langle U^*(t)\, U(t+\tau)\rangle$ is the temporal coherence function. **[1 mark]**

The **coherence time** is defined as $\tau_c = \int_{-\infty}^{\infty} |g(\tau)|^2\, d\tau$. **[0.5 marks]**

For a Lorentzian spectral profile with FWHM $\Delta\nu$, the degree of coherence is an exponential:

$$|g(\tau)| = \exp(-\pi\Delta\nu\, |\tau|) = \exp\left(-\frac{|\tau|}{\tau_c}\right)\,,$$

where $\tau_c = 1/(\pi\Delta\nu)$. **[1 mark]**

The coherence length is

$$l_c = c\,\tau_c = \frac{c}{\pi\Delta\nu}\,.$$

[0.5 marks]

(b) [3 marks]

In a Mach–Zehnder interferometer, the two beams recombine with a relative time delay τ. The output intensity is the superposition:

$$I(\tau) = \langle |U(t) + U(t+\tau)|^2\rangle = \langle |U|^2\rangle + \langle |U|^2\rangle + 2\,\mathrm{Re}[\langle U^*(t) U(t+\tau)\rangle]\,.$$

Using $G(\tau) = G(0)\, g(\tau)$ and writing $g(\tau) = |g(\tau)|\, e^{j2\pi\bar{\nu}\tau}$ (for a symmetric spectrum centred at $\bar{\nu}$):

$$I(\tau) = 2I_0\left[1 + \mathrm{Re}\big(|g(\tau)|\, e^{j2\pi\bar{\nu}\tau}\big)\right] = 2I_0\left[1 + |g(\tau)|\cos(2\pi\bar{\nu}\tau)\right]\,,$$

where $I_0 = \langle |U|^2\rangle/2$ per arm (assuming 50:50 splitting). Absorbing the factor of 2 into the definition of I_0 gives the stated result. **[2 marks]**

The **fringe visibility** is

$$\mathcal{V} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = |g(\tau)|\,.$$

Thus measuring the contrast of the fringe pattern as a function of delay directly yields the modulus of the complex degree of coherence. **[1 mark]**

(c) [5 marks]

The total refractive index in the presence of the Kerr effect is $n(I) = n + n_2 I$, where $I(r) = I_0 \exp(-2r^2/w^2)$.

The phase acquired traversing the slab of thickness d_{g} is

$$\varphi(r) = \frac{2\pi}{\lambda}\, n(I)\, d_{\mathrm{g}} = \frac{2\pi}{\lambda}\left(n + n_2 I_0 e^{-2r^2/w^2}\right) d_{\mathrm{g}}\,.$$

[1 mark]

Near the beam axis ($r \ll w$), expand the exponential:

$$e^{-2r^2/w^2} \approx 1 - \frac{2r^2}{w^2}\,,$$

so the spatially varying part of the phase is

$$\Delta\varphi(r) = \frac{2\pi n_2 I_0 d_{\mathrm{g}}}{\lambda}\left(1 - \frac{2r^2}{w^2}\right).$$

[1 mark]

The r-dependent part is

$$\varphi_{\mathrm{lens}}(r) = -\frac{2\pi n_2 I_0 d_{\mathrm{g}}}{\lambda}\cdot\frac{2r^2}{w^2} = -\frac{4\pi n_2 I_0 d_{\mathrm{g}}}{\lambda w^2}\,r^2\,.$$

[1 mark]

A thin lens of focal length f imparts a quadratic phase $\varphi_{\mathrm{lens}}(r) = -\pi r^2/(\lambda f)$ (from the transmittance $e^{jkr^2/2f}$). Comparing:

$$\frac{\pi}{\lambda f} = \frac{4\pi n_2 I_0 d_{\mathrm{g}}}{\lambda w^2}\,,$$

giving the focal length:

$$\boxed{f = \frac{w^2}{4\, n_2\, I_0\, d_{\mathrm{g}}}}\,.$$

[2 marks] For $n_2 > 0$ this is a converging lens (self-focusing), as expected for the optical Kerr effect.

(d) [6 marks]

Derivation of $n(E_0)$: A DC field E_0 applied to a $\chi^{(2)}$ medium produces a nonlinear polarisation at the optical frequency:

$$P^{(2)}(\omega) = \varepsilon_0 \chi^{(2)}(\omega;\omega,0)\,E(\omega)\,E_0\,.$$

The total displacement field is

$$D = \varepsilon_0\left(1 + \chi^{(1)} + \chi^{(2)}E_0\right)E(\omega)\,,$$

giving an effective refractive index:

$$n(E_0) = \sqrt{1 + \chi^{(1)} + \chi^{(2)}E_0} \approx n_0 + \frac{\chi^{(2)}}{2n_0}\,E_0\,,$$

where $n_0 = \sqrt{1+\chi^{(1)}}$ and the approximation holds for $\chi^{(2)}E_0 \ll n_0^2$. **[2 marks]**

Pockels cell between crossed polarisers: Consider light polarised at 45° to the crystal's principal axes. The two polarisation components propagate with refractive indices

$n_x = n_0 + \Delta n/2$ and $n_y = n_0 - \Delta n/2$, where $\Delta n \propto \chi^{(2)} E_0$ is the electro-optically induced birefringence. [**1 mark**]

After propagating through the crystal of length L, the accumulated phase difference between the two components is

$$\Gamma = \frac{2\pi}{\lambda}\,\Delta n\, L = \frac{2\pi}{\lambda}\frac{\chi^{(2)}}{n_0}\,E_0\,L\,.$$

For a voltage V applied across a crystal of thickness d_c (transverse configuration, $E_0 = V/d_c$), or along the propagation direction (longitudinal, $E_0 = V/L$). Taking the longitudinal case: $\Gamma = (2\pi/\lambda)(\chi^{(2)}/n_0)\,V$. [**1 mark**]

Through crossed polarisers (analyser at 90° to input), the transmitted intensity is

$$I_{\text{out}} = I_{\text{in}} \sin^2\!\left(\frac{\Gamma}{2}\right) = I_{\text{in}} \sin^2\!\left(\frac{\pi V}{2V_\pi}\right)\,,$$

where the **half-wave voltage** V_π is defined as the voltage required for $\Gamma = \pi$ (full transmission):

$$V_\pi = \frac{\lambda\, n_0}{2\chi^{(2)}} \qquad \text{(longitudinal geometry)}\,.$$

[**2 marks**] for the intensity expression and identification of V_π.

Total marks: Section A = 16, Section B = 34, Paper total = 50.